

VISION READING GROUP

Activity Report: Real-Time Tracking Demo & Literature Review

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Reading Group Themes Covered	Real-time bounding-box tracking (OpenCV); Game AI / RL from pixels; Probabilistic deep learning for medical image analysis; Active vision & robotics
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Part 1 — Hands-On Demo: Bounding Boxes & Real-Time Tracking in OpenCV

To connect the reading-group papers to something concrete, I built a small OpenCV script that generates a short synthetic video of three moving objects, then runs a live-style processing loop — background subtraction, contour-based detection, and centroid tracking — exactly the pipeline described in Papers 2 and 3 below. This section shows the actual output rather than a mock-up.

How the pipeline works

```
bg_subtractor = cv2.createBackgroundSubtractorMOG2(varThreshold=25)

for frame in video_stream:
    blurred = cv2.GaussianBlur(frame, (5,5), 0)
    fg_mask = bg_subtractor.apply(blurred)
    fg_mask = cv2.morphologyEx(fg_mask, cv2.MORPH_OPEN, kernel)
    fg_mask = cv2.morphologyEx(fg_mask, cv2.MORPH_CLOSE, kernel)

    contours, _ = cv2.findContours(fg_mask, cv2.RETR_EXTERNAL, ...)
    boxes = [cv2.boundingRect(c) for c in contours if area(c) > 150]

    for box in boxes:
        track_id = match_to_nearest_existing_track(box.centroid)
        draw_rectangle(frame, box)
        draw_label(frame, f'ID {track_id}')
```

Output — Foreground Mask (Background Subtraction Stage)

The mask below is the direct output of `cv2.createBackgroundSubtractorMOG2()` after morphological cleanup. Each moving circle leaves a crescent-shaped blob because MOG2 flags the difference between the object's current and previous position against the learned background model — this is the raw signal the detection stage works from.

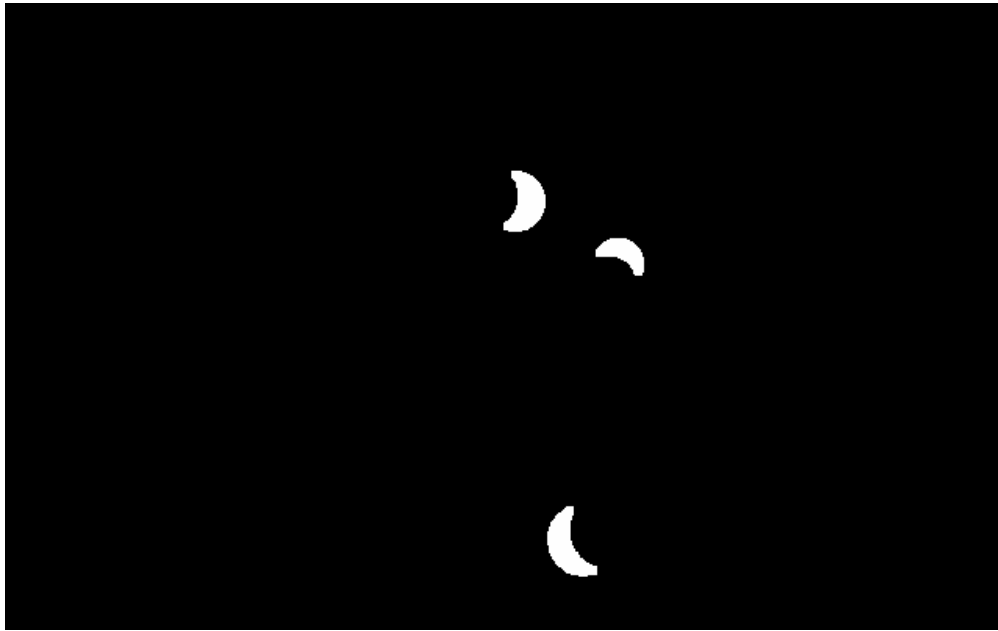


Fig. 1 — Foreground mask from MOG2 background subtraction (own output, generated for this report)

Output — Bounding Boxes with Persistent Tracking IDs

After contour extraction and area filtering, each detected blob is matched to the nearest track from the previous frame using centroid distance (the same idea used by SORT, Paper 2). Objects keep the same ID (ID 2, ID 3, ID 5) as they move, which is what allows a system to count objects once, and not once per frame.

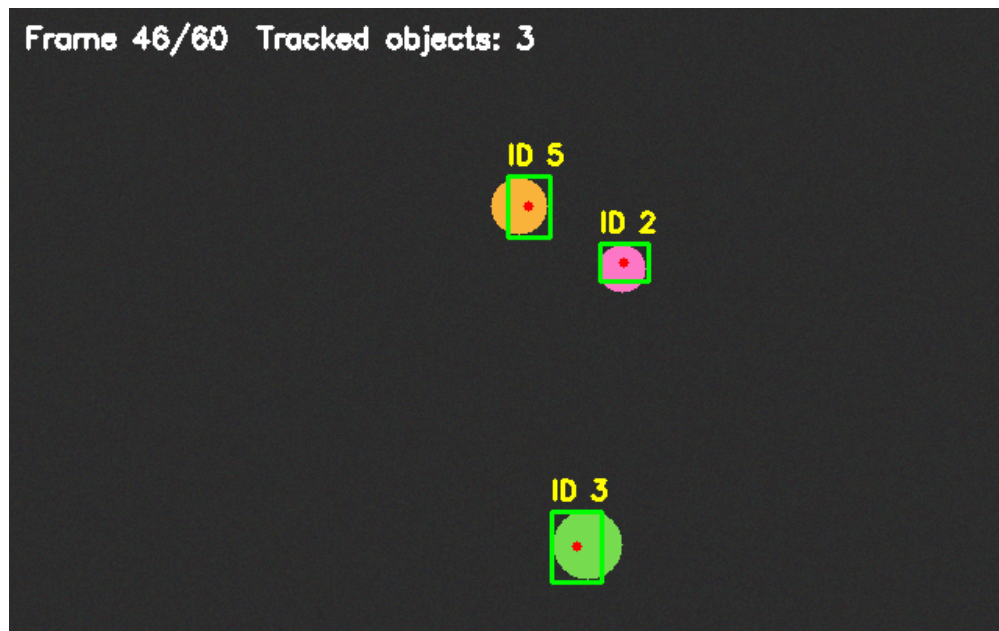


Fig. 2 — Bounding boxes and persistent track IDs on a live-style frame (own output, generated for this report)

Takeaway: even this lightweight, classical pipeline (MOG2 + contours + centroid matching) reproduces the core mechanics behind the modern deep-learning trackers discussed in the papers below — it just swaps a learned detector for background subtraction, and a learned appearance metric for simple distance.

Part 2 — Literature Review (5 Papers)

The five papers below map onto the reading-group's stated focus areas: real-time bounding-box tracking, active vision for robotics and autonomous vehicles, game AI, and probabilistic/uncertainty-aware deep learning for medical image analysis.

Paper 1: Bewley et al. — "Simple Online and Realtime Tracking" (SORT)

Authors: Alex Bewley, Zongyuan Ge, Lionel Ott, Fabio Ramos, Ben Upcroft

Venue / Year: IEEE International Conference on Image Processing (ICIP), 2016

Theme: Real-time bounding-box tracking

Summary

This paper tackles multi-object tracking with an emphasis on speed rather than sophistication. Rather than building a complex appearance model, the authors pair a per-frame object detector with a Kalman filter (to predict where each tracked object should be next) and the Hungarian algorithm (to optimally assign new detections to existing tracks by bounding-box overlap).

The headline result is that detection quality, not tracking cleverness, is the dominant factor in tracking accuracy — improving the underlying detector improved tracking performance by as much as 18.9%. Despite its simplicity, SORT ran at roughly 260 Hz, more than 20 times faster than competing trackers at the time, while remaining competitive in accuracy.

Relevance to the Reading Group Themes

SORT is essentially the algorithmic backbone of the centroid/Kalman-style tracking demonstrated in Part 1 above — matching new bounding boxes to existing tracks frame-to-frame is exactly SORT's core data-association step, just using distance instead of a learned Kalman motion model.

Paper 2: YOLO-based real-time object detection for autonomous vehicles

Authors: Multiple independent studies applying the YOLO family (YOLOv3–YOLOv8) to driving datasets, incl. BDD100K

Venue / Year: Various — IJECE, ResearchGate preprints, 2024–2026

Theme: Active vision / autonomous vehicles

Summary

These studies apply YOLO-family single-pass detectors to road scenes, predicting object bounding boxes and class scores for vehicles, pedestrians, and traffic signs in one forward pass through a CNN, rather than the multi-stage region-proposal pipeline used by older detectors like Faster R-CNN.

Several of the papers propose hybrid designs — e.g., combining YOLO's speed with Faster R-CNN-style RoI pooling, or adding depth-based post-filtering — specifically to cut down false detections near the vehicle

("ghost objects" from shadows or reflections) while keeping inference fast enough for real-time driving decisions.

Relevance to the Reading Group Themes

This is the clearest real-world instance of the 'equip autonomous vehicles with real-time visual perception' theme: bounding-box detection at video frame-rate is the perceptual front end that every downstream driving decision depends on, and it is the same detection stage that feeds a tracker like SORT.

Paper 3: Kendall & Gal — "What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?"

Authors: Alex Kendall, Yarin Gal

Venue / Year: Advances in Neural Information Processing Systems (NeurIPS), 2017

Theme: Probabilistic deep learning for medical image analysis

Summary

This paper formalizes a distinction that much later medical-imaging work builds on: aleatoric uncertainty (irreducible noise inherent in the data or sensor) versus epistemic uncertainty (uncertainty about the model itself, which can shrink with more data). It shows how to have a single neural network output both a prediction and a calibrated estimate of each type of uncertainty, using Monte Carlo dropout for epistemic uncertainty and a learned per-pixel variance term for aleatoric uncertainty.

Applied to semantic segmentation and depth regression benchmarks, models that output both types of uncertainty are shown to be better calibrated and more robust than deterministic point-estimate models — a property that matters far more in safety-critical settings than in typical benchmark leaderboards.

Relevance to the Reading Group Themes

This paper is foundational for the reading group's medical-imaging theme: probabilistic scan analysis for diagnosing neurological conditions is, in practice, largely about knowing when a CNN's diagnosis is unreliable — exactly the aleatoric/epistemic decomposition this paper introduced, which is now standard in Bayesian medical segmentation work (e.g., Bayesian SegNet, probabilistic U-Net).

Paper 4: Mnih et al. — "Human-Level Control Through Deep Reinforcement Learning" (DQN)

Authors: Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. (DeepMind)

Venue / Year: Nature, 2015

Theme: Game AI

Summary

This is the paper that introduced the Deep Q-Network: a single convolutional network that takes raw pixel frames from an Atari 2600 emulator (no hand-engineered game features) and learns, purely from trial-and-error reward signals, to output Q-values for each possible joystick action.

The same network architecture and hyperparameters were used unchanged across 49 different games, and DQN matched or exceeded professional human testers on the majority of them — the first demonstration that a single vision-based RL agent could generalize across a wide range of distinct visual tasks without game-specific tuning.

Relevance to the Reading Group Themes

This is the seminal 'vision → decision' pipeline for game AI: the CNN front end that turns pixels into features is architecturally the same computer-vision component (convolutional feature extraction) used in the detection and tracking papers above — DQN just attaches it to a reinforcement-learning decision head instead of a bounding-box regressor.

Paper 5: "Active Visual Perception: Opportunities and Challenges" (survey)

Authors: Survey covering active-perception systems across retail, autonomous vehicles, sports robotics, and search-and-rescue drones

Venue / Year: arXiv preprint, 2026

Theme: Active vision & robotics

Summary

Unlike passive vision systems that simply process whatever frame a fixed camera happens to capture, active visual perception treats where and how to look as something the system itself controls — repositioning a camera, adjusting a viewpoint, or steering a robot's attention toward the most informative part of a scene.

The survey walks through this idea across several deployed and research settings: autonomous vehicles dynamically reweighting sensor attention toward likely obstacles, retail robots repositioning to check for suspicious behavior, and rescue drones actively scanning disaster sites — tying each back to the classical formulation of "active perception" as controlling sensing to reduce uncertainty about the world.

Relevance to the Reading Group Themes

This paper speaks directly to the reading group's 'equipping autonomous vehicles and field robots with real-time visual perception' theme — it's the conceptual layer above pure detection/tracking, addressing not just what a system sees but how it should move or refocus to see better, which matters for field robots operating in unstructured, changing environments.

Summary Table

Paper	Core Idea	Year	Theme
SORT (Bewley et al.)	Kalman filter + Hungarian algorithm for fast multi-object tracking	2016	Tracking
YOLO for autonomous driving	Single-pass CNN bounding-box detection at video frame-rate	2024–26	Active vision / AV
Kendall & Gal	Aleatoric vs. epistemic uncertainty in Bayesian deep vision models	2017	Probabilistic medical imaging

Mnih et al. (DQN)	Pixels-to-action deep RL, human-level across 49 Atari games	2015	Game AI
Active Visual Perception survey	Controlling sensing/viewpoint, not just passive detection	2026	Active vision / robotics